

12.Implement Iris Flower Classification using KNN

Aim

To implement **Iris Flower Classification using K-Nearest Neighbours (KNN)** in Python and classify iris flowers into their respective species.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Separate the input features and target classes.
4. Split the dataset into training and testing data.
5. Create a KNN classifier with a suitable value of K.
6. Train the model using the training data.
7. Predict the species of the test data.
8. Calculate the accuracy of the model.
9. Display the classification results.

PROGRAM:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

knn = KNeighborsClassifier(n_neighbors=3)

knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Actual Classes:", y_test)
print("Predicted Classes:", y_pred)
print("Accuracy:", accuracy * 100, "%")
```

```

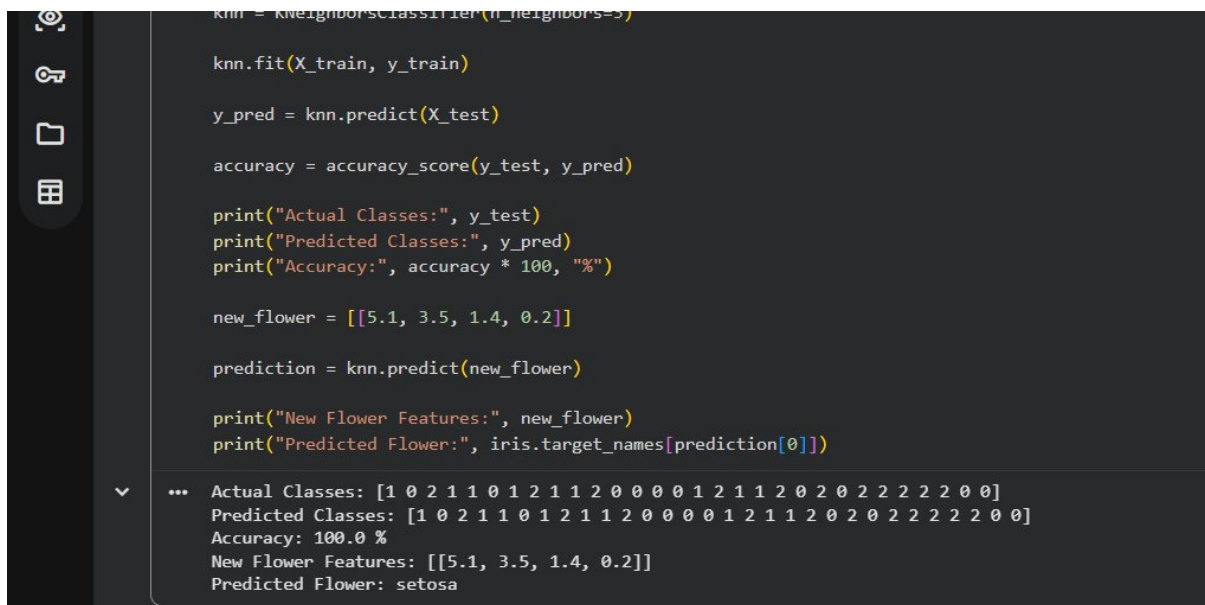
new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = knn.predict(new_flower)

print("New Flower Features:", new_flower)
print("Predicted Flower:", iris.target_names[prediction[0]])

```

OUTPUT:



```

knn = KNeighborsClassifier(n_neighbors=5)

knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Actual Classes:", y_test)
print("Predicted Classes:", y_pred)
print("Accuracy:", accuracy * 100, "%")

new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = knn.predict(new_flower)

print("New Flower Features:", new_flower)
print("Predicted Flower:", iris.target_names[prediction[0]])

```

... Actual Classes: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
 Predicted Classes: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
 Accuracy: 100.0 %
 New Flower Features: [[5.1, 3.5, 1.4, 0.2]]
 Predicted Flower: setosa

RESULT:

Thus, **Iris Flower Classification using KNN** was successfully implemented in Python, and the iris flower species was correctly classified.